

CE 303: Water Resources Engineering

Course Code	CE 303
Course Name	Water Resources Engineering
L-T-P-C	3-0-0-3
Prerequisites	CE 251 - Hydraulic Engineering
Intended for	B.Tech. (Civil Engineering)
Mutual Exclusion	None
Approval	9th Senate (May 2016)
Revised	52nd BoA (Nov 2023) - syllabus revised

Preamble

This course is an advance course of civil engineering which will cover the aspects of hydrology and surface water, watershed management and other hydraulic engineering concepts. This course will unveil the concepts to more advanced level. The effect of hydrological cycle to runoff generation, calculation of runoff and estimation in a catchment. The subject will also provide the learning opportunity to the students to understand the floods and water shed management in hilly terrain. Students will also understand the concept of groundwater flow and transport problems. Learning outcomes of this course are anticipated as follows:

- Students will understand the basics of hydrology, meteorology, rainfall-runoff calculations and factors affecting the rainfall and runoff relation.
- Students will learn the concepts of storm hydrology, s-curve and depth area duration curves for estimation of rainfall over a catchment area
- Students will develop the skills to analyze floods, flood routing systems flood forecasting, return period of any flood, and probability analysis.

Course Modules

- **Module I: Introduction:** Hydrological Cycle, Water budget equation, Reynold Transport Theory, Principle of mass, momentum, and energy balance in Hydrology (6 Hours)
- **Module II: Precipitation:** Atmospheric Circulation, Water vapor, Precipitation measurement, rain gauge network, Mean Precipitation, Maximum Intensity/Depth Duration Curve, Rainfall patterns in India and Himachal Pradesh (8 Hours)
- **Module III: Abstractions from Precipitation:** Evaporation and its estimation, Evapotranspiration, Estimation of Evapotranspiration, Potential Evapotranspiration, Interception, Infiltration (6 Hours)
- **Module IV: Hydrometry:** Measurement of Stage, Velocity, Area Velocity Method, Dilution Technique, Electromagnetic method, Ultrasonic Method, Indirect method, Rating Curve (6 Hours)
- **Module V: Storm Hydrology:** Hydrographs, unit hydrograph theory, S-curve, Mass and flow duration curve, depth area duration curve, runoff estimation. (8 Hours)
- **Module VI: Floods and Droughts:** Flood estimation, Frequency analysis, Risk and Reliability, Hydrologic and Hydraulic routing, Clark's and Nash's model (6 Hours)
- **Module VII: Groundwater Hydrology:** Types of Aquifer and characteristics, Application of Darcy Law, Groundwater flow equations, well hydraulics. (2 Hours)

Text Books

1. K. Subramaniya, *Engineering Hydrology*, Tata MacGraw Hill, New Delhi, 2013.
2. V.T. Chow, D.R. Maidment, and L.W. Mays, *Applied Hydrology*, McGraw Hill, 1988.

References

1. H.M. Raghunath, *Hydrology – Principles, Analysis and Design*, Wiley Eastern Ltd., 2006.
2. V.P. Singh, *Elementary Hydrology*, Prentice Hall, 1993.
3. R.K. Linsley, J.B. Franzini, D. L. Freyberg and G. Tchobanoglous, *Water Resource Engineering* 4th Edn., McGraw Hill Book Co., 1992.